

Nile Egypt Labs

Website Rebuild — Architecture & User Journeys

Technical architecture, visitor journey, admin journey, content model, integrations and project boundaries

Project Goal

- Rebuild the public Nile Egypt Labs website with a modern, responsive interface.
- Use the existing Results Portal color palette and visual theme as the brand reference, while creating a cleaner and more modern website experience.
- Provide an admin dashboard so the owner/team can manage offers, YouTube videos, laboratory devices/equipment and branch information without developer intervention.
- Keep the existing patient-results portal as a separate application; the new site links to it rather than rebuilding it.

1. High-Level Architecture

VISITOR	PUBLIC WEBSITE	CONTENT / API	DATABASE / STORAGE
Mobile / Tablet / Desktop	Next.js + React + TypeScript Tailwind + RTL	Next.js server/API Zod validation	Supabase PostgreSQL Supabase Storage

Admin Dashboard +
Secure Authentication

Separate Results Portal Boundary

- Online Results button opens the existing Nile Egypt Labs Results Portal.
- Patient laboratory results, authentication and portal APIs are outside this website rebuild scope.
- Any modification, security assessment, API integration or redevelopment of the Results Portal is a separate project.

2. Technology Stack — Layers

L1 — Presentation

- Next.js, React, TypeScript, Tailwind CSS, reusable UI components, responsive layouts, Arabic/RTL support.

L2 — Application / API

- Next.js Server Components/Server Actions/API routes as appropriate; Zod for server-side validation; secure form handling.

L3 — Authentication & Authorization

- Supabase Auth; admin-only protected dashboard; server-side authorization; secure sessions/cookies.

L4 — Data

- Supabase PostgreSQL for structured website content: offers, videos, devices, branches, contact/appointment submissions and settings.

L5 — Media

- Supabase Storage for website images and device/offer media. YouTube remains the video host; videos are embedded by URL/Video ID.

L6 — Integrations

- Existing Results Portal link; YouTube embeds; WhatsApp Business contact actions; email notification service if required.

L7 — Security

- HTTPS/TLS, HSTS, CSP, X-Content-Type-Options, Referrer-Policy, Permissions-Policy, input validation, rate limiting, secure file handling and dependency scanning.

L8 — Deployment & Monitoring

- GitHub, CI/CD, Vercel production/staging environments, environment variables, error monitoring and basic analytics.

3. Visitor Journey

01	Visit website Homepage loads with responsive design and the Results Portal visual theme.
02	Explore Visitor browses services, about information, offers, videos, devices/equipment and branch information.
03	Offers Visitor opens current offers and sees only published/active offers.
04	Videos Visitor watches selected YouTube videos directly inside the website.
05	Devices Visitor explores newly added laboratory devices/equipment and related media.
06	Contact Visitor taps Contact and can choose the appropriate contact action.
07	WhatsApp WhatsApp Business opens with a predefined message, e.g. a greeting/inquiry template.
08	Online Results Visitor taps Online Results and is redirected to the existing Results Portal.

4. Admin Journey

01	Admin Login Admin opens /admin and authenticates securely.
02	Dashboard Admin sees quick access to offers, videos, devices/equipment, branches and website content.
03	Manage Offers Create/edit/publish/unpublish offers; add image, title, description and validity dates.
04	Manage Videos Paste YouTube URL/Video ID; set title, description, thumbnail/metadata, publish status and display order.
05	Manage Devices Add new laboratory device/equipment with image, description, related YouTube video and publish status.
06	Manage Branches Maintain the full branch directory, with a dedicated Head Office/Main Branch flag so the head office is clearly identified among other branches.
07	Manage Content Update approved website content/settings within the agreed CMS scope.
08	Publish Changes are saved in the database and reflected on the public site according to publish status.

5. Content Model

Content Type	Admin Fields	Public Usage
Offers	Title, description, image, validity, status, order	Offers section / homepage highlights
Videos	YouTube URL/ID, title, description, status, order	Embedded YouTube player
Devices	Name, image, description, related video, status, order	New Devices / Equipment section
Branches	Name, type/head-office flag, address, phone, hours, map data	Branches / Head Office presentation
Contact	WhatsApp number + predefined message/configuration	Contact CTA → WhatsApp Business

WhatsApp Contact Behavior

- A Contact/WhatsApp CTA can open the client's WhatsApp Business number.
- The URL can include a predefined message so the conversation starts with a controlled inquiry template.
- Example message: "Hello Nile Egypt Labs, I would like to ask about your services/offers." The exact final message and number are supplied/approved by the client.

Head Office vs. Other Branches

- The website should support multiple sub-branches.
- The client specifically wants the Head Office/Main Branch to be clearly identifiable because of competitors/market positioning.
- Therefore the data model should not assume there is only one branch; it should support multiple branches with a Head Office/Main Branch designation.

6. Security & Operational Boundaries

Website Security

- HTTPS from launch; HSTS; modern security headers; server-side validation; rate limiting; secure admin authentication and authorization; protected admin routes; safe media upload rules; dependency updates and scanning.
- Do not expose admin permissions only through frontend controls.
- Do not store patient laboratory results in the new website database.

Existing Audit Findings Driving the Rebuild

- The current marketing site has no working HTTPS.
- The public site uses very old jQuery 1.10.2 and Bootstrap 3.3.4.
- Security headers are absent on the current hosts.
- Existing forms do not have a functioning submission backend.
- The existing site contains unfinished template content.

Results Portal

- Existing results portal remains a separate application.
- The new site provides a clear Online Results entry point.
- Testing or changing the patient-results application requires separate written scope and authorization.